

BOROD, RomaniaWMO: **150950**Lat: **46.9936N**Lon: **22.5903E**Elev: **1096**StdP: **14.12**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	15.4	5.4	7.6	13.3	9.6	9.4	17.5	21.6	37.9	18.1	37.9	2.0	120	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.4	88.3	69.6	85.4	68.8	82.3	67.7	71.8	84.2	70.3	82.4	69.0	80.4	5.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	106.7	77.5	66.3	100.7	75.6	64.8	95.6	74.5	36.1	84.7	34.9	82.6	33.7	80.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	14.2	11.5	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	30.3	36.2	41.3	50.6	59.5	66.1	67.8	68.9	60.5	51.6	44.5	34.0		
	DBStd	15.04	8.78	8.00	7.63	8.43	6.08	5.76	5.23	5.61	7.75	7.52	7.99	8.05		
	HDD50	2162	611	387	282	95	8	0	0	0	6	74	202	497		
	HDD65	5513	1075	805	734	431	188	56	32	26	176	415	615	960		
	CDD50	2535	0	2	13	114	301	483	552	586	320	125	37	2		
	CDD65	414	0	0	0	1	16	89	119	148	40	1	0	0		
	CDH74	4086	0	0	0	31	180	850	1111	1505	378	31	0	0		
	CDH80	1220	0	0	0	1	18	208	328	571	94	0	0	0		
(14) Wind	WSAvg	3.7	3.1	3.9	4.6	4.7	4.0	3.7	3.9	3.2	3.3	3.3	3.4	3.4		
(15) Precipitation	PrecAvg	38.7	2.2	2.0	2.4	2.9	3.8	4.8	4.7	3.2	3.6	3.1	2.8	3.0		
	PrecMax	52.0	4.8	5.2	6.9	4.6	6.4	7.8	8.2	8.2	9.9	8.1	6.5	6.5		
	PrecMin	27.8	0.9	0.4	0.7	0.7	1.2	1.3	2.0	0.2	0.6	0.2	0.0	0.2		
	PrecStd	6.5	1.0	1.2	1.6	1.0	1.4	1.7	2.0	1.8	2.1	1.8	1.3	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	59.4	67.2	77.5	82.4	87.9	90.6	93.2	87.3	77.3	68.5	58.2		
		MCWB	47.5	49.5	53.8	60.0	65.5	71.8	69.8	70.7	67.7	61.1	55.5	48.8		
	2%	DB	48.1	54.8	61.9	73.7	79.0	85.4	87.3	89.4	82.4	73.9	64.6	51.6		
		MCWB	44.0	46.9	50.3	58.2	64.4	69.8	69.8	69.0	65.9	59.1	53.9	45.4		
	5%	DB	44.7	51.6	57.7	69.6	75.6	82.1	83.4	86.0	78.5	69.4	60.5	47.9		
		MCWB	41.3	44.9	48.4	56.4	63.1	68.6	68.1	68.9	64.0	58.8	52.4	44.0		
	10%	DB	42.1	48.2	53.5	65.3	72.2	79.0	80.2	82.2	74.0	64.4	56.4	45.2		
		MCWB	39.7	43.3	45.7	53.9	61.0	67.4	67.0	67.8	62.6	56.2	50.3	42.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	50.8	55.2	62.3	67.5	74.6	72.3	73.3	70.2	64.0	56.7	49.4		
		MCDB	53.0	57.2	65.3	75.4	77.5	84.5	84.4	89.0	84.2	71.1	67.9	55.6		
	2%	WB	44.6	47.7	51.6	59.6	65.8	71.9	71.0	71.6	67.3	61.3	54.9	47.2		
		MCDB	47.4	53.2	60.0	71.4	76.2	82.2	82.9	85.4	78.5	70.5	63.0	51.7		
	5%	WB	42.3	46.0	49.0	57.3	64.0	69.8	69.7	70.1	65.2	59.1	53.1	44.4		
		MCDB	44.8	50.7	56.8	68.0	73.5	80.5	81.2	83.1	75.7	68.1	59.9	47.9		
	10%	WB	39.6	44.1	46.6	55.0	62.1	68.0	68.2	68.6	63.4	57.0	51.1	42.2		
		MCDB	41.9	48.1	52.6	63.8	70.5	77.9	79.0	80.5	72.2	63.2	56.3	45.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	13.7	16.8	19.8	19.4	19.6	20.5	21.4	18.9	18.5	14.8	10.5		
		MCDBR	14.9	19.0	23.0	26.1	25.3	23.9	25.6	27.4	25.7	26.7	20.7	14.6		
		MCWBR	10.4	12.1	13.4	13.2	13.4	11.2	11.3	11.3	11.5	14.5	11.6	10.2		
	5% WB	MCDBR	14.1	16.6	21.5	24.0	23.0	22.3	22.7	24.9	22.0	24.2	19.2	13.9		
		MCWBR	10.0	11.0	13.0	12.6	12.4	11.3	10.6	11.1	10.6	13.5	10.9	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.356	0.401	0.455	0.437	0.434	0.453	0.455	0.417	0.382	0.353	0.327		
	taud		2.291	2.235	2.159	2.044	2.160	2.208	2.165	2.153	2.221	2.295	2.320	2.325		
	Ebn at Noon		235	251	257	253	263	265	257	251	249	239	222	220		
	Edh at Noon		25	32	40	50	46	44	45	44	38	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		357	614	994	1397	1697	1863	1828	1681	1182	787	442	273		
	RadStd		51	86	146	144	167	139	161	155	138	96	62	39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		+1.26	N/A	N/A	+1.85	N/A	N/A	-229	-344	+198	+99

Nomenclature: See separate page